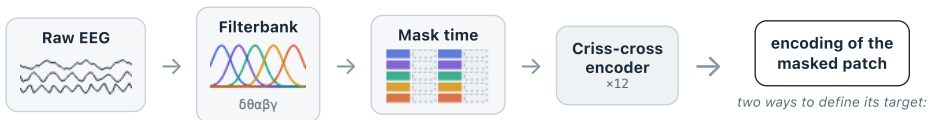


# What to predict for a masked patch — and why band power, not the band signal?

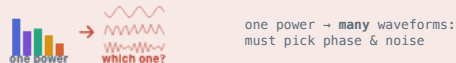


## OURS Predict band power — from context



⇒ the encoder must learn the **spectral state** (= the label), not the waveform

## ALTERNATIVE Reconstruct the band signal



⇒ reconstruction: capacity spent fitting phase & noise (CBraMod's failure)

Band-power prediction is only the **pretext task**: after pretraining we **discard the head** and keep the **encoder** — the foundation model that transfers to sleep / depression / workload. Nothing is reconstructed at inference.